

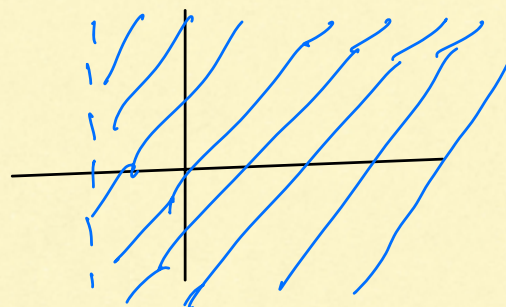
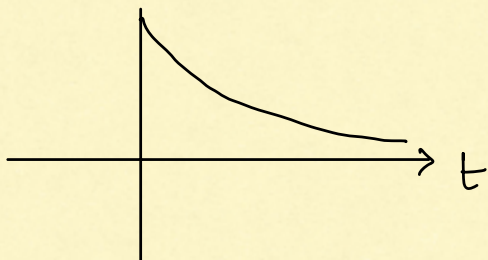
Lecture - 36

Laplace Transform

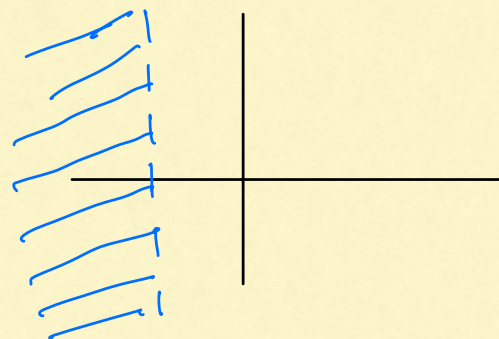
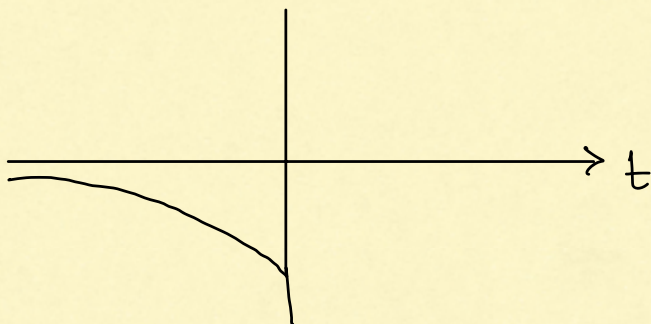
Chapter 9 of Oppenheim, Willsky and Nawab

Recap:

$$\textcircled{1} \quad e^{-at} u(t) \xleftrightarrow{L} \frac{1}{s+a}, \quad \operatorname{Re}(s) > -\operatorname{Re}(a)$$



$$\textcircled{2} \quad -e^{-at} u(-t) \xleftrightarrow{L} \frac{1}{s+a}, \quad \operatorname{Re}(s) < -\operatorname{Re}(a)$$



③ If $x_1(t) \xleftrightarrow{L} X_1(s)$, ROC R_1

$x_2(t) \xleftrightarrow{L} X_2(s)$, ROC R_2

then:

$x_1(t) + x_2(t) \xleftrightarrow{L} X_1(s) + X_2(s)$,

ROC contains $R_1 \cap R_2$.

Depending on the specific problem:

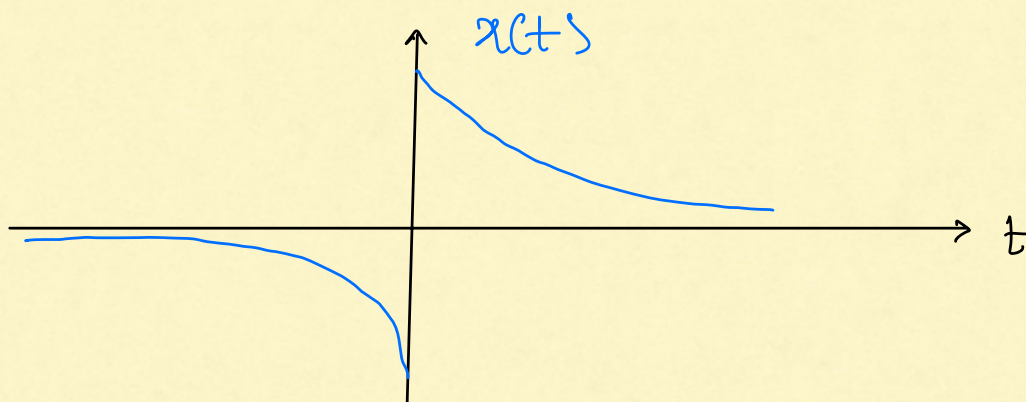
$ROC = R_1 \cap R_2$

or ROC is larger than $R_1 \cap R_2$.

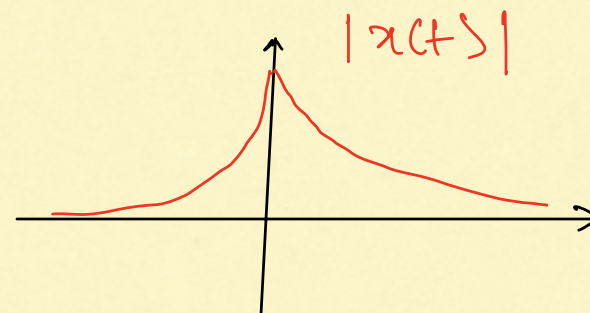
① Examples: Continued.

$$⑥ \quad x(t) = e^{-t}u(t) - e^{2t}u(-t)$$

$$= \begin{cases} e^{-t} & \text{if } t > 0 \\ -e^{2t} & \text{if } t < 0 \end{cases}$$



$$\Rightarrow |x(t)| = \begin{cases} e^{-t}, & t > 0 \\ e^{2t}, & t < 0 \end{cases}$$

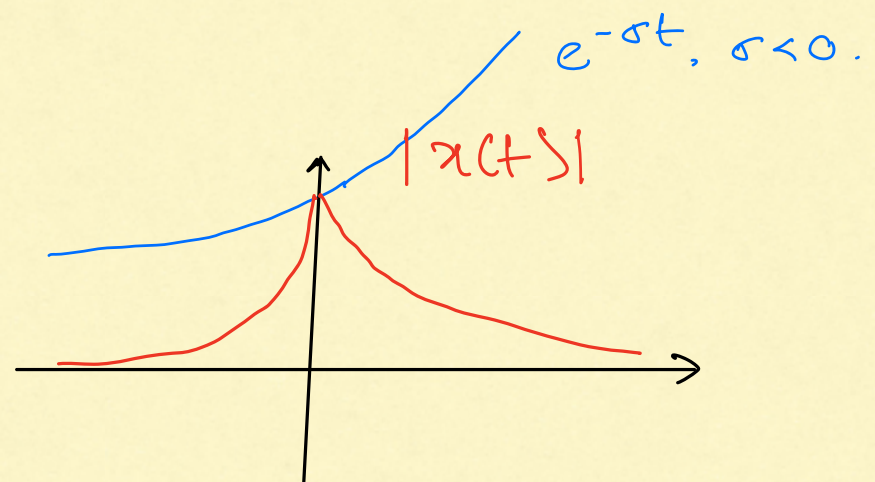
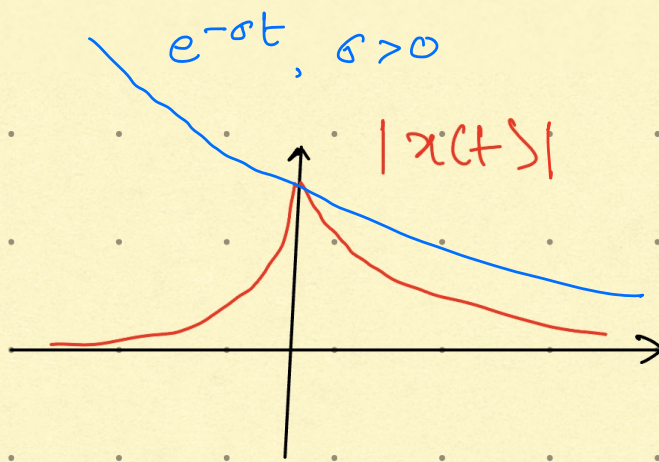


(a) Roc: say $s = \sigma + j\omega$

$$\int_{-\infty}^{+\infty} |x(t) \cdot e^{-st}| dt = \int_{-\infty}^{+\infty} |x(t)| \cdot |e^{-(\sigma + j\omega)t}| dt.$$

$$= \int_{-\infty}^{+\infty} |x(t)| \cdot |e^{-\sigma t}| \cdot |e^{-j\omega t}| dt$$

$$= \int_{-\infty}^{+\infty} |x(t)| \cdot e^{-\sigma t} dt.$$



$$\int_{-\infty}^{+\infty} |x(t)| e^{-\sigma t} dt = \int_0^{\infty} |x(t)| \cdot e^{-\sigma t} dt + \int_{-\infty}^0 |x(t)| \cdot e^{-\sigma t} dt.$$

$$= \int_0^{\infty} |e^{-t} u(t)| \cdot e^{-\sigma t} dt + \int_{-\infty}^0 |-e^{2t} u(-t)| e^{-\sigma t} dt$$

↓

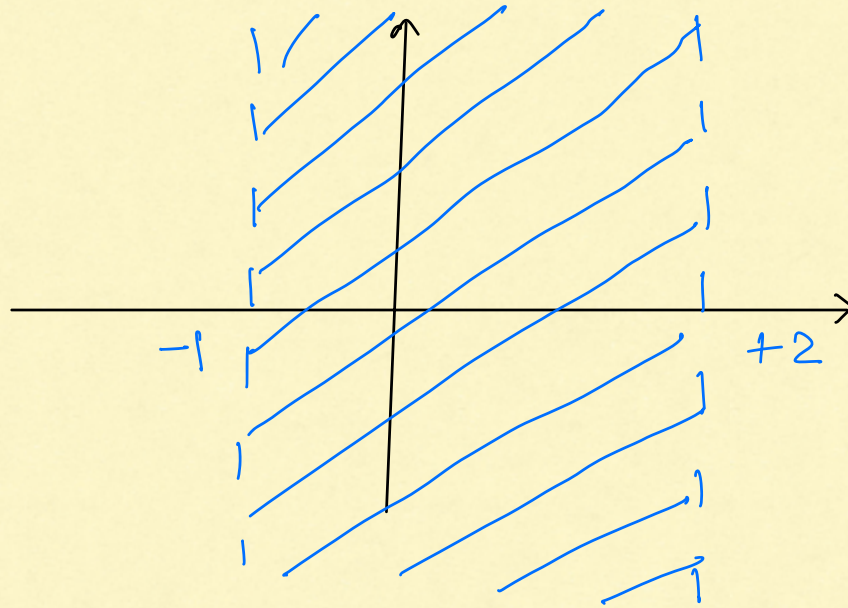
↓

We need both the integrals to be finite.

$\therefore s \in \text{ROC} \iff s \in \text{ROC of } e^{-t} u(t) \text{ and } s \in \text{ROC of } -e^{2t} u(-t).$

$\text{ROC} = \text{intersection of ROCs of } e^{-t} u(t), -e^{2t} u(-t).$

$$\therefore \text{ROC} = \{s: \operatorname{Re}(s) > -1\} \cap \{s: \operatorname{Re}(s) < 2\}.$$



$$-1 < \operatorname{Re}(s) < +2$$

(b) Value of $X(s)$:

If $s \in \text{ROC}$,

$$X(s) = \int_{-\infty}^{+\infty} x(t) \cdot e^{-st} dt,$$

$$X(s) = \int_{-\infty}^{+\infty} [e^{-t}u(t) - e^{2t}u(-t)] e^{-st} dt$$

$$= \int_{-\infty}^{+\infty} e^{-t}u(t) \cdot e^{-st} dt + \int_{-\infty}^{+\infty} -e^{2t}u(-t) e^{-st} dt$$

$$= \frac{1}{s+1} + \frac{1}{s-2}$$

$$= \frac{2s-1}{(s+1)(s-2)} = \frac{2 \cdot (s-1/2)}{(s+1)(s-2)}$$

Pole - Zero Plot

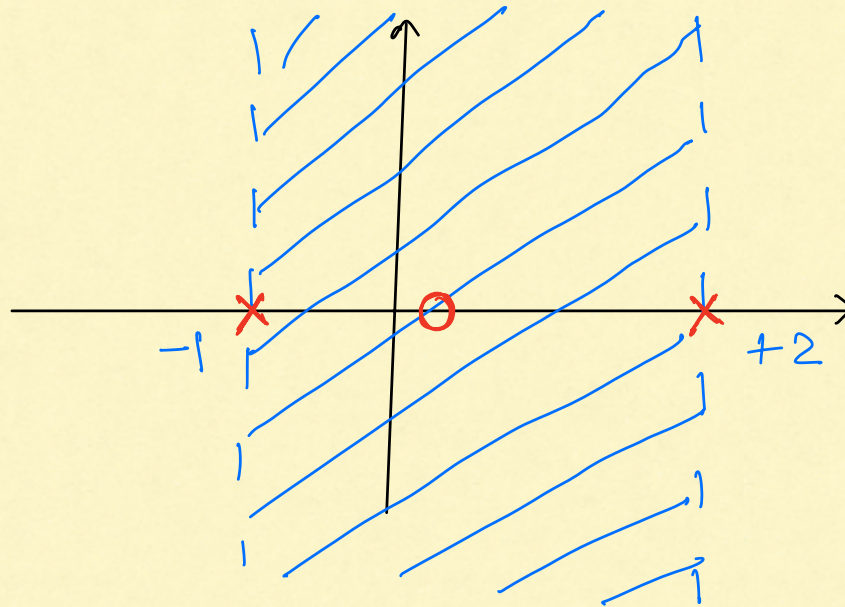
If $X(s)$ is rational, i.e., $X(s) = \frac{N(s)}{D(s)}$ $\xrightarrow{\text{Polynomials in } s}$

Assume no common factors between $N(s)$, $D(s)$.

- Roots of $N(s)$: **Zeros** of $X(s)$, use o
- Roots of $D(s)$: **Poles** of $X(s)$, use x

Example:
$$\frac{2 \cdot (s - 1/2)}{(s+1)(s-2)}$$

Pole-Zero Plot



$$e^{-t}u(t) - e^{2t}u(-t) \xleftrightarrow{L} \frac{2(s - \frac{1}{2})}{(s+1)(s-2)}, \quad -1 < \operatorname{Re}(s) < +2$$

II

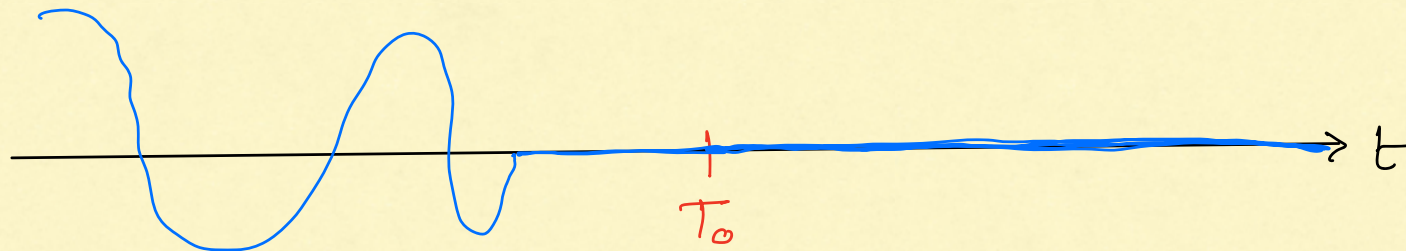
Left-sided and Right Sided Signals

Left-sided Signal

$x(t)$ is left-sided if $\exists$ a T_0 such that

$$x(t) = 0 \text{ for all } t > T_0$$

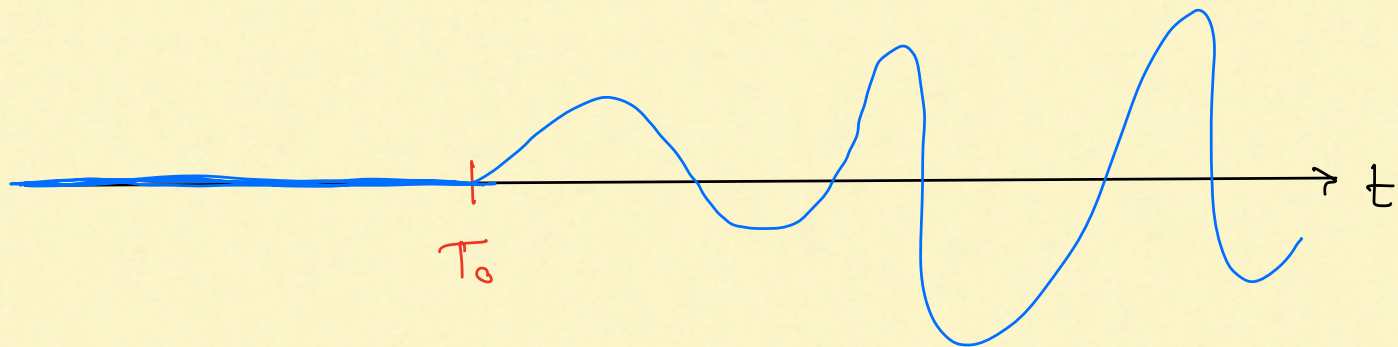
(T_0 can be any constant, positive or negative)



Right-Sided Signal

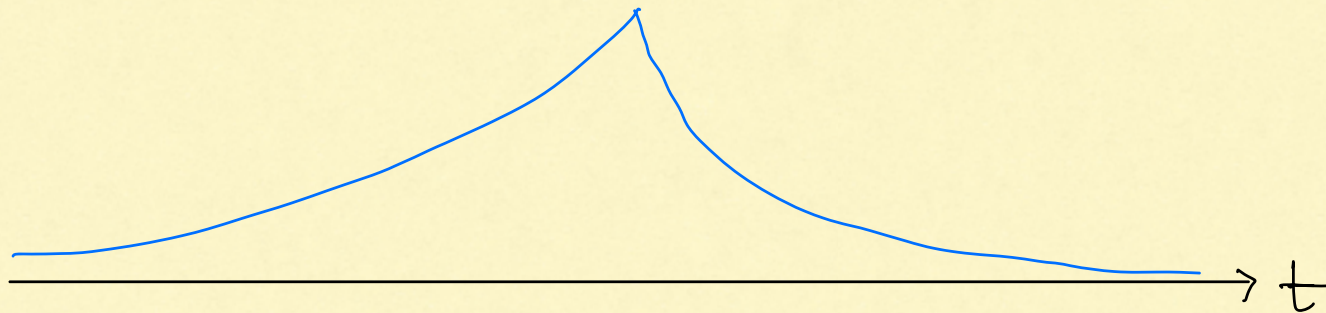
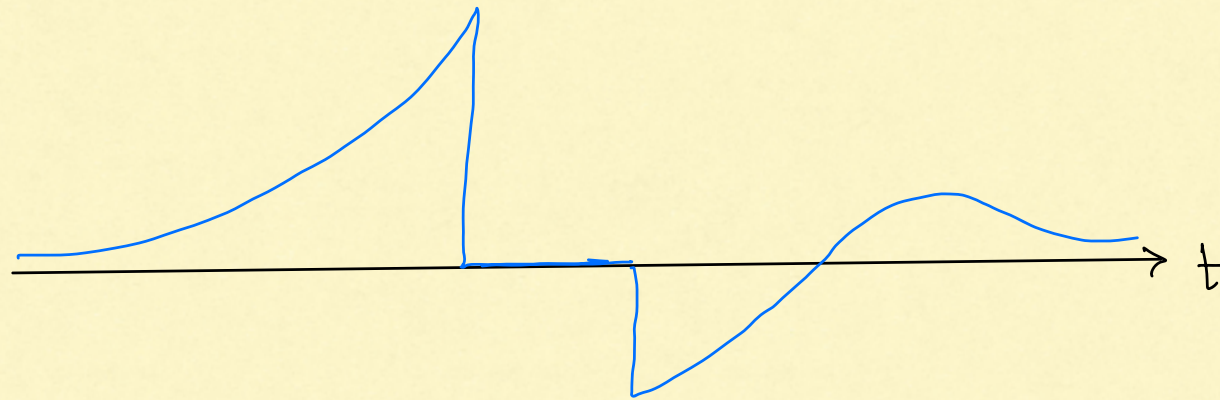
$x(t)$ is right-sided if there exists a T_0 such that

$$x(t) = 0 \text{ for all } t < T_0$$



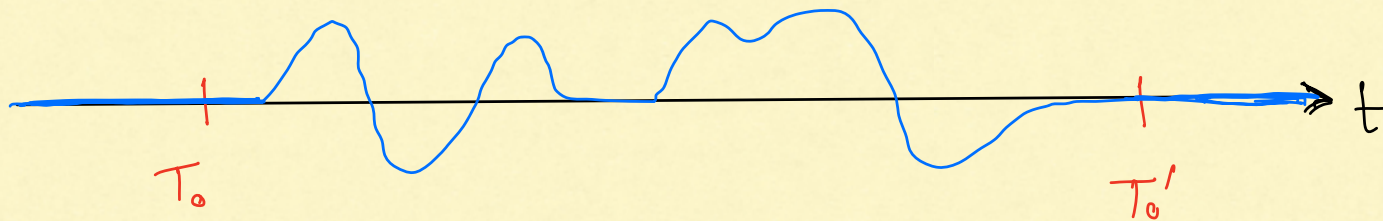
Two-Sided Signal

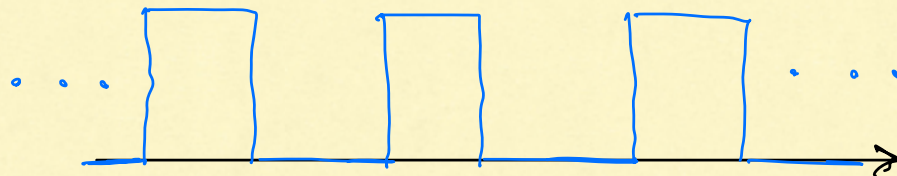
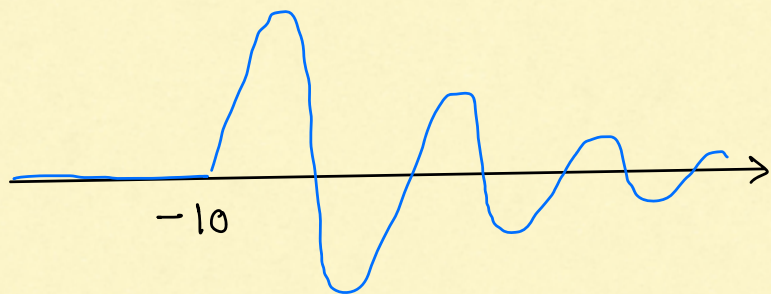
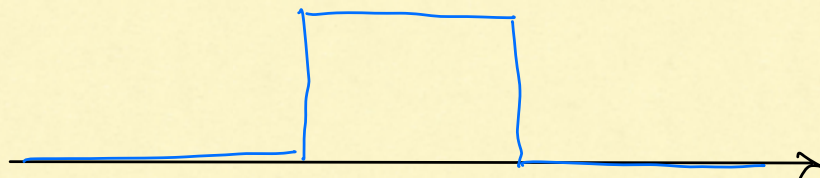
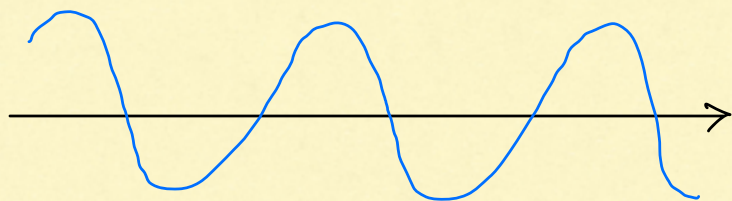
$x(t)$ is neither left- nor right-sided.



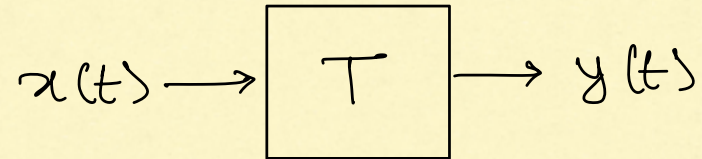
Finite-Duration Signals

$x(t)$ is both left- and right-sided.





Causal Systems



A system is causal if the value of the output at t depends only on present and/or past inputs.

i.e., $y(t)$ is independent of $x(\tau)$, $\tau > t$.
for all values of t .

Systems that are not causal : non-causal systems.

Examples:

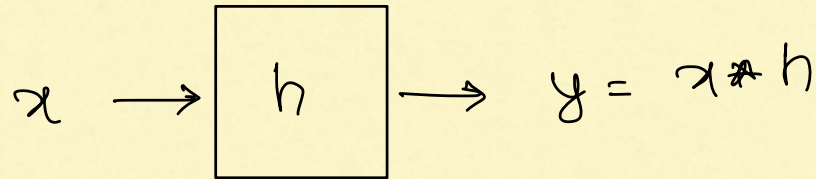
$$\textcircled{1} \quad y(t) = \int_{-\infty}^t x(\tau) d\tau$$

$$\textcircled{2} \quad y(t) = \int_{t-1/2}^{t+1/2} x(\tau) d\tau$$

$$\textcircled{3} \quad y(t) = x(-t)$$

$$\textcircled{4} \quad y(t) = x(2t)$$

Causal LTI Systems

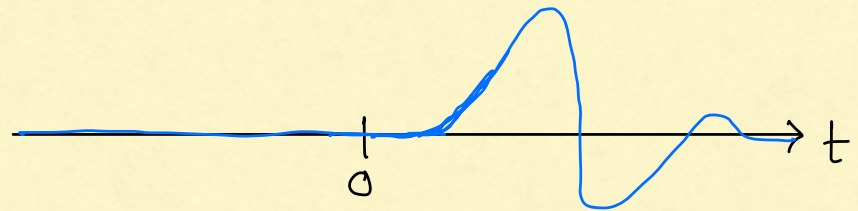
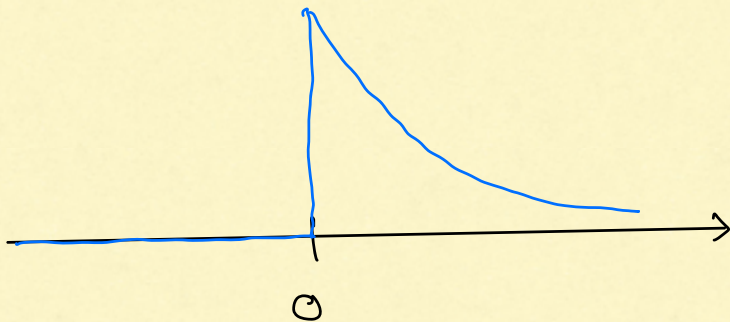
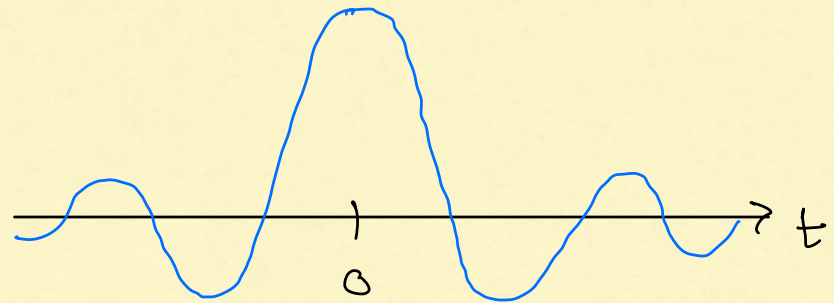
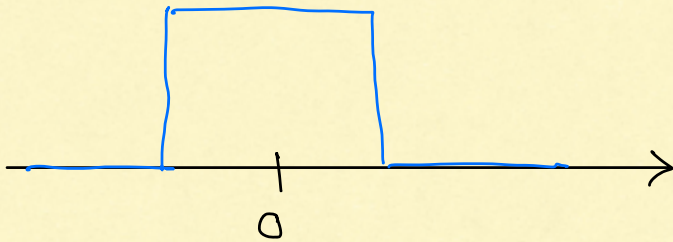


$$y(t) = \int_{-\infty}^{+\infty} h(\tau) x(t-\tau) d\tau$$

$$= \int_{-\infty}^0 h(\tau) \underbrace{x(t-\tau)}_{\substack{\text{future} \\ \text{values of } x}} d\tau + \int_0^{\infty} h(\tau) \underbrace{x(t-\tau)}_{\substack{\text{past values} \\ \text{of } x}} d\tau$$

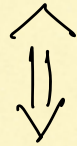
An LTI system is causal if and only if

$$h(\tau) = 0 \text{ for all } \tau < 0$$



Note:

LTI system is causal



$$h(t) = 0 \quad \forall t < 0$$



$h(t)$ is a right-sided signal.

Converse not true!

